

Suryachandra Prasad Pasupuleti

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ABOUT ME

As a first-year PhD student in Computer Science at the University of York, I am currently working on the exciting challenge of developing energy-aware acceleration techniques for High-Performance Computing (HPC) workloads. My research aims to explore innovative approaches that balance computational performance with power efficiency, enabling more sustainable and efficient computing architectures.

Prior to embarking on my PhD journey, I spent two years as an FPGA Engineer at Covnetics Limited in Nuneaton, where I worked on a couple of high-profile projects related to aviation and communication systems. This experience not only deepened my understanding of digital circuit design but also instilled in me a passion for harnessing the power of FPGAs to accelerate complex computing tasks.

I am now leveraging this expertise to tackle the pressing challenge of energy-aware acceleration, seeking to develop novel techniques that can significantly reduce the power consumption of HPC systems while maintaining or even exceeding their performance capabilities. Through my research, I aim to contribute meaningfully to the development of more sustainable and efficient computing technologies that can have a lasting impact on various fields, from scientific simulations to real-time data processing.

CORE SKILLS

- Proficient in FPGA design using various tools from prominent vendors, including Xilinx Vitis and Intel Quartus.
- Strong expertise in VHDL, with a solid understanding of RTL (Register-Transistor Level) design languages.
- Skilled in C and C++ programming languages, with experience in python scripting and Tcl.
- Strong coding, debugging, and testing skills, with experience in collaborating on FPGA projects and ensuring seamless integration of hardware and software components.
- Hands-on experience with RTL design debugging, utilizing simulators such as ModelSim and Cocotb framework for testing and verification of digital circuits.
- Expertise in hardware/software co-design and acceleration, with knowledge of FPGA-based design methodologies and system-on-chip (SOC) architecture optimization for efficient code mapping and deployment.

EDUCATION

University of York <i>PhD in Computer Science</i>	York, United Kingdom <i>Sept 2024 – present</i>
University of York <i>MSc in Digital Systems Engineering</i>	York, United Kingdom <i>Sept 2021 – Sept 2022</i>
Birla Institute of Technology and Science Pilani <i>Bachelor of Engineering in Electrical and Electronics, M.S in Chemistry</i>	Goa, India <i>Aug 2014 – Aug 2019</i>

WORK EXPERIENCE

Covnetics Limited <i>FPGA Design Consultant</i>	Nuneaton, United Kingdom <i>Oct 2022 – July 2024</i>
• Successfully deployed deep learning inference solutions on the AMD KV260 System-on-Chip (SoC) board • Contributed to an avionics project that required the development of a real-time video processing system on Intel Stratix FPGA for multiple video feed streams. Additionally, I successfully integrated a NIOS II processor into the FPGA design flow.	

- Developed and implemented an FPGA-based gateway device for high-speed communications systems, utilizing the Intel Cyclone FPGA device to emulate a comprehensive network interface. As part of this project, I contributed to designing and developing a protocol conversion solution that captures and converts STM-1 and STM-4 data streams into ATM packets, and vice versa. This enabled seamless integration with existing ATM networks, while also allowing for efficient conversion between STM and ATM protocols. Automated verification using Tcl and Python scripts, developing comprehensive testbenches for rapid and reliable testing.

Research Intern, RISE lab

April 2020 – June 2020

Indian Institute of Technology Madras

Chennai, India

- As part of our project, I utilized the Cocotb framework to develop and execute test cases for the various RTL modules in SHAKTI RISC-V processor.

Research Intern, Computer Systems Group

Aug 2019 – Dec 2019

International Institute of Information Technology Hyderabad

Hyderabad, India

- I successfully implemented an end-to-end image processing system on Zedboard and Zynq Ultrascale FPGA boards, showcasing the feasibility of high-performance computing on these platforms.
- I utilized the Petalinux platform to develop device drivers for the image processing system, ensuring optimal performance and communication between the hardware and software components.
- I successfully demonstrated the proof-of-concept of the lab-developed framework in real-time image processing applications, verifying its effectiveness and applicability in industry-relevant scenarios.”

ACADEMIC EXPERIENCE

Research Assistant

Nov 2024 – March 2025

SYSTRON Lab, University of York

York, United Kingdom

- Worked on calculating an estimated power consumption for 6G using the analytical model.
- The project is funded by Communications Hub for Empowering Distributed Cloud Computing Applications and Research (CHEDDAR).

Bachelor Thesis

June 2018 – Dec 2018

International Institute of Information Technology Hyderabad

Hyderabad, India

- Successfully completed my Bachelor thesis Title: Accelerating Image processing Pipelines on FPGAs.
- I optimized the implementation of Bilateral filter and Bilateral Grid algorithms for FPGA, exploiting parallelism at multiple levels to achieve improved performance.

CERTIFICATES

- AMD Adaptive Computing Partner Program
- Comprehensive VHDL course by Doulos